# LedgerForge - Phase 3 Submission Report

Sudharsana | Python Track | LedgerForge | Phase 3 - Connected System

Repository: github.com/Sudharsana27/LedgerForge | Git tag: phase-3 | 23 October 2026

## Executive summary

Phase 3 extends the existing FastAPI and SQLAlchemy ledger with authenticated access, auditable payroll and expense workflows, versioned tax configuration, and financial statements. Earlier employee, salary, attendance, leave, payroll-calculation, payslip, invoicing, and double-entry ledger behavior remains covered by regression tests.

## Five-task completion status

| Task | Status | Evidence / limitation |
|---|---|---|
| 11 - Expenses | Complete | 501 source bills reconcile; approval, duplicate, and payment tests pass. |
| 12 - Payroll | Partial | Lifecycle / posting / reversal tests pass; independent Check 2 is blocked by missing inputs. |
| 13 - Security | Implemented | JWT, role isolation, expiry, inactive-user tests; local secret must be strengthened. |
| 14 - Tax | Partial | Versioning, audit, TDS and PF/ESI reports tested; source rates are training samples. |
| 15 - Reports | Complete | Official trial balance: 40 accounts, 0 mismatches; P&L; matches and balance-sheet difference is 0.00. |

## Authentication and authorization

API access uses 30-minute signed bearer tokens. Password hashes are verified with bcrypt, user activation is checked on every request, and employee-facing routes enforce record ownership. ADMIN and HR roles protect operational and financial surfaces; only ADMIN may version tax slabs. Cross-employee payslip access is HR-only. Login is disabled until a unique SECRET_KEY of at least 32 bytes is configured. The source user CSV has no credentials, so passwords are provisioned interactively with scripts/set_user_password.py and are never committed.

## Payroll lifecycle

Payroll runs are created as DRAFT from explicit per-employee variable-pay, TDS, and other-deduction inputs. Idempotency keys are bound to saved request contents; processing replays each item through the calculation engine and checks run totals. Approval and its journal posting are atomic and require a different user from the run creator. Payment records net-pay disbursement; reversal creates balanced reversal entries for both the payroll posting and any recorded payment. Repeated posting, payment, and reversal operations are idempotent.

# Financial, expense, and tax controls

## Expense workflow

Expense bills progress through SUBMITTED, APPROVED, and PAID. The workflow prevents self-approval, detects duplicate vendor / reference / date / amount combinations, validates posting accounts, and records balanced accrual and settlement journals. Spend reconciliation compares monthly and vendor-month totals against the supplied expenses.csv source.

## Financial reporting

Trial balance, profit-and-loss, and balance-sheet reports include posted journal activity only. Contra-asset normal balances and unclosed current-period profit are represented so the balance sheet remains balanced. The trial-balance export is formatted for comparison by the official LedgerForge checker.

## Tax and statutory reporting

Tax slab changes create new effective-dated versions and audit records instead of rewriting previously saved payroll. TDS estimates use configured slabs, rebate threshold, deduction, and cess. Statutory reports sum saved payroll items for the selected period. The API flags sample/training tax data; these figures are not approved production tax policy or tax advice.

## Source-data constraints

The supplied data has 18 payroll runs and 7,340 items but no independent payroll_inputs.csv. The independent batch validator therefore processed 7,340 rows, skipped all 7,340 for missing variable-pay / TDS / other-deduction inputs, and calculated zero successful comparisons. Reusing the reference payroll totals as calculated output would not be an independent check and was deliberately avoided.

Tax and statutory rate rows are explicitly marked sample or training data. The existing source also documents conflicting ESI eligibility and two one-paisa overtime rounding differences. The calculation engine preserves its explicit input and rate rules; it does not alter historical values to force a match.

## Role-permission summary

| Role | Permitted scope |
|---|---|
| Admin | Employee/finance/payroll operations, reports, and tax configuration; own payslip only. |
| HR | Organization employee/payroll/finance records, reports, and payslips. |
| Employee | Own employee, salary, attendance, leave, and payslip records only. |
| Public | Health check and login only. |

# Validation and release record

## Automated and official checks

| Validation | Result |
|---|---|
| Pytest regression + Phase 3 | 177 passed; 1 warning |
| Workflow and tax coverage | Payroll journal posting, payment and reversal; historical payroll preservation; PF/ESI/TDS reports |
| Official source checks | 5/5 passed |
| Official trial balance | 40 accounts; 0 mismatches |
| Expense reconciliation | 501 bills, 18 months; 0 monthly or vendor-month mismatches |
| Migration | Seeded database backed up, adopted at e7c2d10f4b39, upgraded to a390f1b6d2e8; SQLite integrity check OK |

## Official vs. LedgerForge totals

| Measure | Official | LedgerForge |
|---|---|---|
| Revenue | 1,260,930,052.00 | 1,260,930,052.00 |
| Expenses | 1,036,246,464.42 | 1,036,246,464.42 |
| Profit | 224,683,587.58 | 224,683,587.58 |

Official payroll summary: 18 runs; employer cost 861,586,392.42. Balance sheet at 2026-09-30: assets 463,762,958.65; liabilities 77,499,607.07; equity 161,579,764.00; current profit 224,683,587.58; difference 0.00.

## Operational notes

The additive Phase 3 migration is revision a390f1b6d2e8. Fresh databases use alembic upgrade head. The supplied seeded SQLite database had a stale legacy Alembic marker; its schema was verified and backed up before stamping the known baseline and applying the additive migration. Downgrade is intentionally blocked to protect financial history and credentials.

Before enabling login, configure a strong local SECRET_KEY and provision active users with the interactive password utility. Before treating tax or historical payroll as production-ready, obtain approved statutory policy and independent payroll input records. All supplied company data is fictional training data.

## Problems and resolutions

The seeded database had a stale migration marker and older columns. A verified backup, schema preflight, safe baseline stamp, and additive migration preserved its data. Missing independent payroll inputs caused the 7,340 rows to be reported as skipped rather than filled with invented amounts. The configured JWT secret is 25 bytes; authentication correctly fails closed until it is rotated to at least 32 bytes.

## Learning and Phase 4

Phase 3 applied least-privilege authorization and separated payroll/expense approval from posting. It reinforced the value of atomic, idempotent journal operations and effective-dated configuration. Reconciliation must use independent inputs, not reference results. Phase 4 should add approved payroll inputs and statutory policy, then harden deployment, monitoring, backups, and operational recovery.